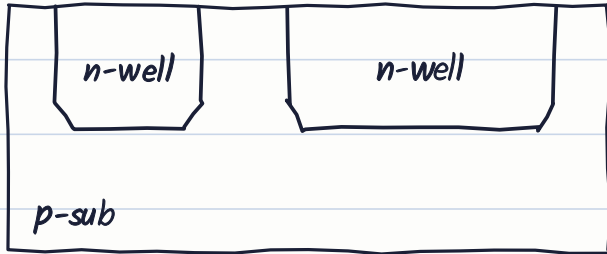


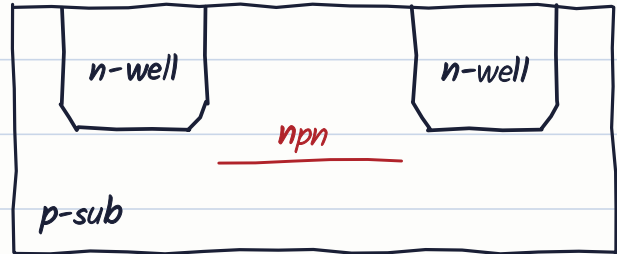
VLSI HW 1

Q1

upper cut



lower cut



pn junction? YES, at every n-well / p-sub boundary

parasitic BJT? YES, lateral npn between neighboring n-wells

(n-well = E, p-sub = B, n-well = C), see lower cut

Q2

Ex 2.5: $R_{sq} = 5 \text{ k}\Omega$, $C_{sq} = 5 \text{ fF}$ (per 10 by 10 square)

$l = 2000/10 = 200$ squares

$$t_d = 0.35 R_{sq} C_{sq} l^2 = 0.35(5 \text{ k}\Omega)(5 \text{ fF})(200)^2$$

$$t_d = 0.35(1 \text{ M}\Omega)(1 \text{ pF})$$

$$t_d \approx 350 \text{ ns}$$

Q3

$$C_{j0} = 25 \text{ aF} / (0.5 \text{ }\mu\text{m})^2 = 100 \text{ aF}/\mu\text{m}^2 = 10^{-4} \text{ F/m}^2$$

$$C_j = C_{j0} / \sqrt{1 + V_R/V_{bi}} \quad (m = 0.5, V_{bi} = 1)$$

$$\text{beginning (1 V): } 10^{-4}/\sqrt{2} = 7.07 \times 10^{-5} \text{ F/m}^2$$

$$\text{middle (0.5 V): } 10^{-4}/\sqrt{1.5} = 8.16 \times 10^{-5} \text{ F/m}^2$$

$$\text{end (0 V): } 10^{-4}/\sqrt{1} = 1 \times 10^{-4} \text{ F/m}^2$$

Q4

$$R = (1 \text{ k}\Omega)(200/2) = 100 \text{ k}\Omega$$

$$C = (400 \text{ }\mu\text{m}^2)(100 \text{ aF}) + (404 \text{ }\mu\text{m})(50 \text{ aF}) = 40 + 20.2 = 60.2 \text{ fF}$$

$$t_d = 0.35(100 \text{ k}\Omega)(60.2 \text{ fF})$$

$$t_d \approx 2.1 \text{ ns}$$

Q5

$$l = 10,000/50 = 200 \text{ squares, so } R = 1 \text{ M}\Omega \text{ (same)}$$

each 50 by 50 square has 25x the area of a 10 by 10

$$C_{sq} = 25(5 \text{ fF}) = 125 \text{ fF, } C = 200(125 \text{ fF}) = 25 \text{ pF}$$

depletion cap goes UP (25x more area, same F/m^2)

delay goes UP: R same, C is 25x bigger

$$t_d = 0.35(1 \text{ M}\Omega)(25 \text{ pF}) \approx 8.75 \text{ }\mu\text{s} \text{ (25x slower)}$$